🐍 Getting Started with Python

📌 1. Introduction to Programming

- **Programming** = giving step-by-step instructions to a computer to accomplish a task.
- Instructions are written using a **programming language**.
- The set of instructions is called a **program**.
- Computers follow instructions literally, fast, and precisely — they need exact logic.

💡 2. Why We Learn Programming

- Develops **logical thinking** & **problem-solving skills** (computational thinking).
- Automate repetitive tasks → save time & effort.
- Gateway to careers in **AI, Data Science, Web Dev, App Dev, Cybersecurity**.
- Builds creativity: games, websites, automation tools, and more.
- Mandatory in CBSE curriculum to nurture future-ready skills.

🌐 3. Programming Languages & Classification

- **Low-level languages:** Machine language (0/1), Assembly language – hardware dependent.
- **High-level languages:** Python, Java, C++, JavaScript – human readable, portable.

📊 4. Familiar / Trending Languages (2025 Context)

Language	Primary Use
🐍 Python	AI, ML, Data Science, Web, Automation, Education
☕ Java	Android apps, Enterprise systems
⚙️ C++	Game dev, OS, performance-critical apps
🌐 JavaScript	Web development (frontend + backend)
📱 Kotlin / Swift	Android / iOS development
📊 R	Statistics, Data visualization

> ✅ Python is the most recommended first language due to its simple, English-like syntax and vast community support.

🖥️ 5. Language Processors (Compiler, Interpreter, Assembler)

- **Compiler:** Translates entire source code at once into machine code. Example: C, C++.
- **Interpreter:** Translates and executes code line by line. Python uses an interpreter → easier debugging.
- **Assembler:** Converts assembly language to machine code.

💡 Python's reference implementation **CPython** is written in C and acts as an interpreter + compiler hybrid (bytecode).

🐍 6. Basics of Python Programming

🔹 Introduction to Python

- High-level, general-purpose, interpreted language.
- Created by **Guido van Rossum** and first released in **February 1991**.
- Named after the BBC comedy show **"Monty Python's Flying Circus"** (1970s).
- CPython is the standard implementation (written in C).

✨ Features of Python

- Easy to read & learn → beginner friendly.
- Interpreted (no separate compilation).
- Cross-platform (Windows, macOS, Linux).
- Open source & free.
- Dynamically typed (no explicit variable type declaration).
- Rich standard library + thousands of third-party modules.
- Supports procedural, object-oriented, and functional programming.

🌍 Where is Python used in the real world?

- Web development (Django, Flask)
- Data Science & Analytics (Pandas, NumPy)
- Artificial Intelligence / Machine Learning (TensorFlow, PyTorch)
- Scientific computing, Automation & Scripting
- Cybersecurity, Game development (Pygame)
- Finance, quantitative trading

📱 Applications developed using Python

- **Instagram** (backend)
- **YouTube** (core components)
- **Spotify** (backend & analytics)
- **Netflix** – recommendation system
- **Dropbox** – desktop client
- **Reddit**, **Uber** (dynamic pricing & algorithms)

⚡ 7. Execution Modes in Python (Interactive vs Script Mode)

🟢 Interactive Mode

- Interacting directly with Python Shell (>>> prompt).
- Instant output after each statement.
- Commands are NOT saved automatically as a file.
- Best for testing small pieces of code and quick experiments.

```
>>> print("Hello Python")
Hello Python
>>> 15 + 30
45
```

📁 Script Mode

- Write multiple lines of code in a **.py** file and execute the entire program.
- Efficient for large programs, reusable code.
- Output appears in console or terminal.

```
# saved as first.py
name = "Class 11"
print("Welcome to Python,", name)
```

👋 8. Executing a Simple "Hello World" Program

In Interactive Mode:

```
>>> print("Hello World")
Hello World
```

In Script Mode:

- Open any IDE/text editor (IDLE, VS Code, Thonny).
- Type: `print("Hello World")`
- Save as `hello.py` → run using terminal: `python hello.py`
- Output: `Hello World`

🛠️ 9. Where to Write / Run Python Code?

- **Python IDLE** – IDE (built-in).
- **Anaconda Navigator** – Package manager with Jupyter Notebook, Spyder.
- **Jupyter Notebook** – Web-based interactive environment (great for teaching).
- **Spyder** – Scientific IDE.
- **PyCharm** – Professional IDE.
- **Thonny** – Perfect for absolute beginners.
- **VS Code** – Powerful source-code editor with extensions.
- **Sublime Text** – Lightweight editor.

📓 10. Jupyter Notebook – Key Highlights (CBSE friendly)

- Web-based interactive environment, opens in a browser.
- Code organized into **cells** – runs interactively.
- Shortcut to run a cell: `Shift + Enter`.
- Supports **Markdown cells** (text, images, tables) + Code cells (Python).
- Output (charts, tables, images) appears right below the cell.
- Save notebooks as `.ipynb` files, perfect for sharing and reports.
- Great for combining explanation, code, and results in one document.

🗂️ 11. Quick Revision Table

Concept	Key Point
Python Creator	Guido van Rossum
First Release	February 1991
Name Inspiration	Monty Python's Flying Circus (BBC comedy)
Python is ...	Interpreted, high-level, dynamically typed
Two Modes	Interactive Mode & Script Mode
Interactive Mode	Immediate output, not saved
Script Mode	.py file, run whole program
Language Processor for Python	Interpreter (CPython)
Jupyter run cell shortcut	Shift + Enter
Jupyter file extension	.ipynb